Exercise 4:

a) $X_t = a + t Z_t$

- $E[X_t] = a + t \underbrace{E[Z_t]}_{0}$ (linearity E)

$= a$

- $\forall k \in \mathbb{N}$,

a has no effect on cov so,

$Cov(X_t, X_{t-k}) = Cov(a + t Z_t, a + (t-k) Z_{t-k})$

$= t(t-k)\, Cov(Z_t, Z_{t-k})$

$$= \begin{cases} t^2 \sigma^2, & \text{if } k = 0 \\ 0, & \text{else} \end{cases}$$

$\gamma_0 = t^2 \sigma^2$ depends on t so $\{X_t\}$ is not stationary.

b) $Y_t = a + b Z_t + c Z_{t-1}$

- $E[Y_t] = a + b \underbrace{E[Z_t]}_{0} + c \underbrace{E[Z_{t-1}]}_{0}$

$E[Y_t] = a$ (is a constant)